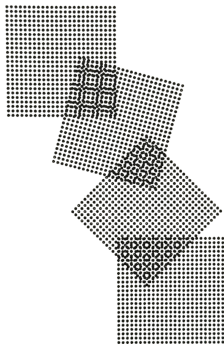


Tutorial: how to build an (F)HE scheme from scratch, using lattices

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Little note: these slides simplify a lot of concepts to explain some intuitions – needed to keep the talk to a high-level. Refer to the actual papers for the precise definitions/results of stuff.

Homomorphic encryption

The limits of public-key encryption

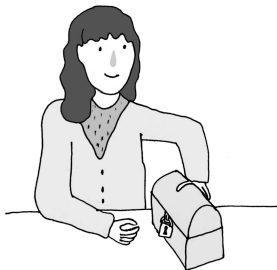
Imagine that Alice wants to send a secret message to Bob...



Homomorphic encryption

The limits of public-key encryption

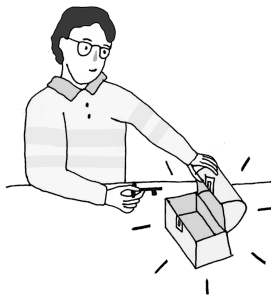
Once locked, it can only be opened by someone who has the key.



Homomorphic encryption

The limits of public-key encryption

Once Bob receives the box, he will be able to open it!



Homomorphic encryption

The limits of public-key encryption

A public key encryption scheme allows Alice and Bob to do that safely:

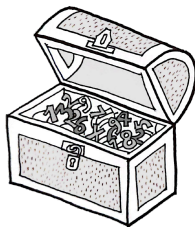
- ▶ The public key pk is an instance of a hard problem (e.g., factorization, discrete logarithm, knapsack problem...)
- ▶ The secret key sk is a trapdoor (i.e., a solution of that problem).

If someone intercepts the conversation, they will have to solve a hard problem to retrieve the message!

Homomorphic encryption

Computing on encrypted data

Imagine a more complex situation, Alice, having her key, puts some values in her box... but she wants someone without the key to be able to evaluate some function!



Homomorphic encryption

Computing on encrypted data

Imagine to turn the (public-key) box into a (homomorphic encryption) glove box: anyone can manipulate the data, but nobody can open the box!



Homomorphic encryption

Computing on encrypted data

When Alice opens the box up, she will find the result!



Homomorphic encryption

Computing on encrypted data

- ▶ A **homomorphic encryption** (HE) scheme is a particular type of cryptosystem where operations can be performed on ciphertexts.
- ▶ Given $\text{enc}(a)$ and $\text{enc}(b)$, it must hold that:

$$\text{dec}(\text{enc}(m_1)) + \text{dec}(\text{enc}(m_2)) = \text{dec}(\text{enc}(m_1 + m_2))$$

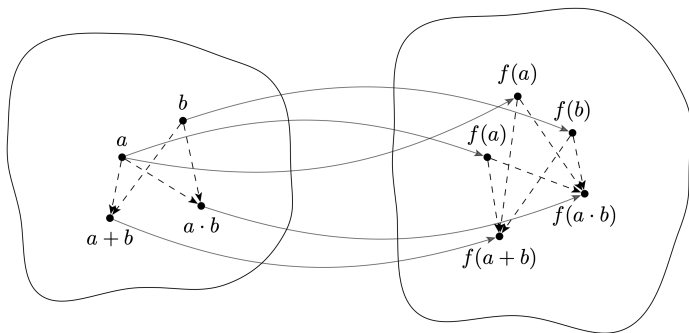
$$\text{dec}(\text{enc}(m_1)) \times \text{dec}(\text{enc}(m_2)) = \text{dec}(\text{enc}(m_1 \times m_2))$$

Homomorphic encryption

Computing on encrypted data

Plaintext space

Ciphertext space



Homomorphic encryption

Computing on encrypted data

We ask for the evaluation of functions almost every time we use our devices

- ▶ Asking to a search engine for a query.
- ▶ Translating a sentence.
- ▶ Downloading emails.
- ▶ Accessing files on the cloud.
- ▶ ...

Homomorphic encryption

Computing on encrypted data

We are (slowly) getting some experimental alternatives! E.g.:

- ▶ <https://spiralwiki.com> allows to access a Wikipedia-like service using FHE.
- ▶ Apple² uses FHE to recover metadata about e-mail senders.

²<https://machinelearning.apple.com/research/homomorphic-encryption>

Lattices

A *lattice* $\mathcal{L}$ is a discrete subset of $\mathbb{R}^n$. Formally it is defined given a *basis* $B \in \text{GL}_n(\mathbb{R})$ and is the set of all integer combinations of the basis vectors:

$$\mathcal{L}(B) = \{B\mathbf{x} : \mathbf{x} \in \mathbb{Z}^n\}$$

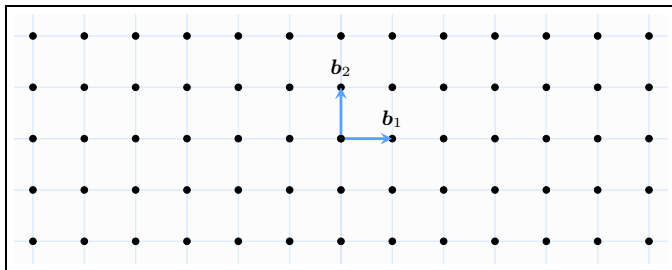
Or, from another perspective, by applying a linear transformation to $\mathbb{Z}^n$:

$$\mathcal{L}(B) = B \cdot \mathbb{Z}^n$$

Lattices

Fundamentals

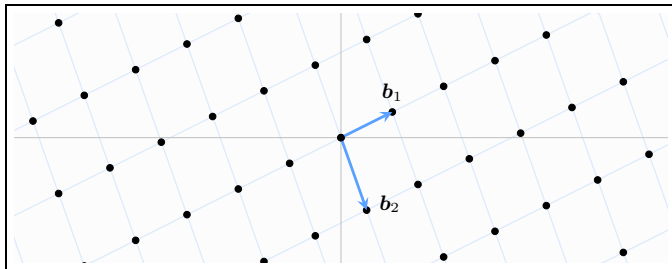
The simplest lattice: $\mathbb{Z}^n$ (with $n = 2$):



Lattices

Fundamentals

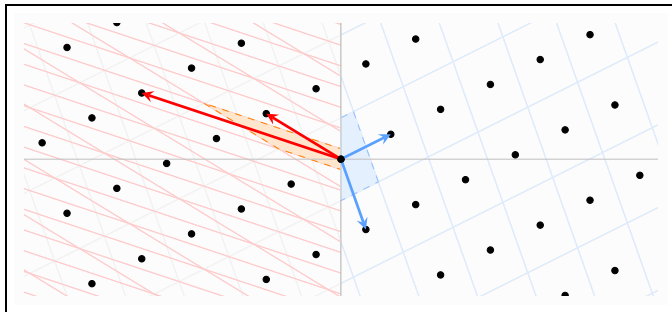
Another lattice, generated by a different set of basis vectors



Lattices

Fundamentals

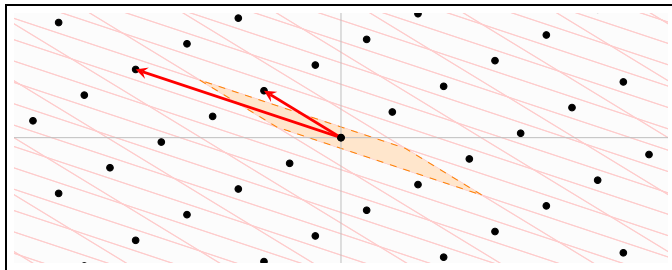
Two lattices $\mathcal{L}(B)$, $\mathcal{L}(B')$ are equivalent if and only if $B' = B \cdot U$ for some unimodular U .



Lattices

Fundamentals

But the *quality* of the tiling depends on the quality of the basis!



Lattices

Fundamentals

To summarize:

- ▶ Two lattices $\mathcal{L}(B)$ and $\mathcal{L}(B')$ are equivalent if and only if $B' = B \cdot U$, where $U \in \text{GL}_n(\mathbb{Z})$ is a unimodular matrix (i.e., with $\det = \pm 1$).
- ▶ Given a good basis, it is easy to transform it in a bad basis (e.g., sample a sufficiently bad unimodular transformation).
- ▶ Given a bad basis, there is no efficient way to transform it in a good basis! Best we can do is with exponential approximation factors [LLL82].

Lattices

q -ary lattice

- In cryptography, we usually deal with q -ary lattices. A lattice $\mathcal{L}$ is called q -ary if

$$q\mathbb{Z}^m \subseteq \mathcal{L} \subseteq \mathbb{Z}^m$$

Row-generated q -ary lattice

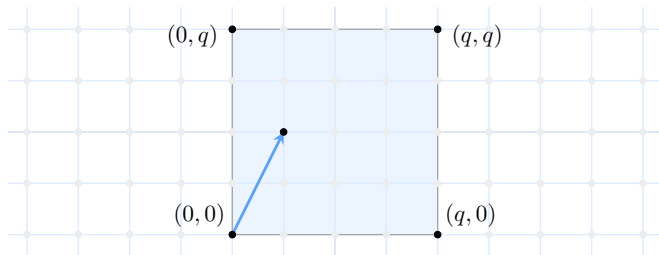
$$\mathcal{L}_q(A) = \{\mathbf{y} \in \mathbb{Z}^m : \mathbf{y} \equiv A\mathbf{x} \bmod q \text{ for some } \mathbf{x} \in \mathbb{Z}_q^n\}$$

for some $A \in \mathbb{Z}_q^{m \times n}$. In other words, $\mathcal{L}_q(A) = A\mathbb{Z}^n + q\mathbb{Z}^m$.

Lattices

q -ary lattice

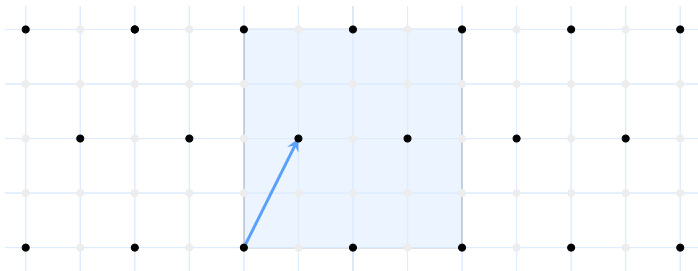
- For instance, fix $m = 2, n = 1, q = 5$ and consider the q -ary basis $A = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ to obtain $\mathcal{L}_q(A)$.



Lattices

q -ary lattice

- For instance, fix $m = 2, n = 1, q = 5$ and consider the q -ary basis $A = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ to obtain $\mathcal{L}_q(A)$.



Lattices

q -ary lattice

- ▶ Another way of instantiating a q -ary lattice is via the “parity check” approach (term borrowed from codes).

Parity check q -ary lattice

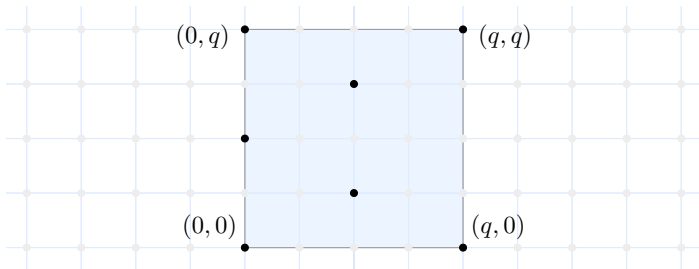
Fix positive integers $n \leq m \leq q$ and $A \in \mathbb{Z}_q^{m \times n}$.

$$\mathcal{L}_q^\perp(A) = \{\mathbf{x} \in \mathbb{Z}^m : A^t \mathbf{x} \equiv \mathbf{0} \pmod{q}\} \subseteq \mathbb{Z}^m$$

Lattices

q -ary lattice

- For instance, fix $m = 2, n = 1, q = 5$ and consider the q -ary basis $A = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ to obtain $\mathcal{L}_q^\perp(A)$:



Lattices

Duality

- ▶ Given a (primal) lattice $\mathcal{L}$, its *dual* lattice $\mathcal{L}^*$ is:

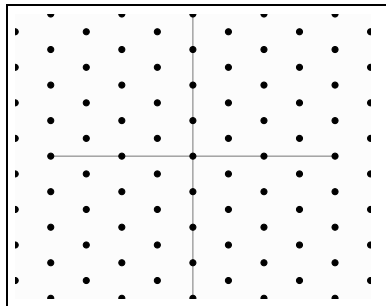
$$\mathcal{L}^* = \{x \in \text{span}(\mathcal{L}) : \langle x, v \rangle \in \mathbb{Z} \text{ for all } v \in \mathcal{L}\}$$

- ▶ It contains all the points in the span of the primal lattice that have integer inner product with all the (primal) lattice points.
- ▶ Given a full-rank basis $B \in \text{GL}_n(\mathbb{R})$, the basis for the dual lattice of $\mathcal{L}(B)$ is simply B^{-t} (i.e., $\mathcal{L}^* = \mathcal{L}(B^{-t})$)

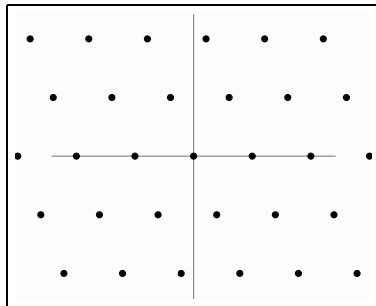
Lattices

Duality

- Observation: the dual lattice can be “partitioned” using vectors in the primal lattice.



(a) Primal lattice $\mathcal{L}$

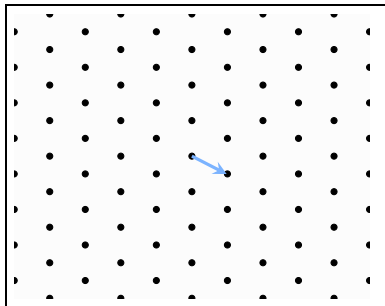


(b) Dual lattice $\mathcal{L}^*$

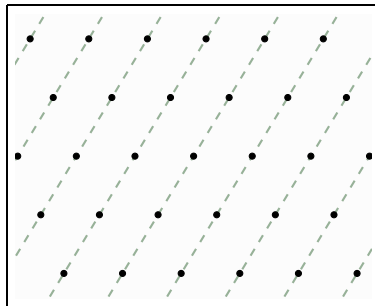
Lattices

Duality

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(a) Primal lattice $\mathcal{L}$



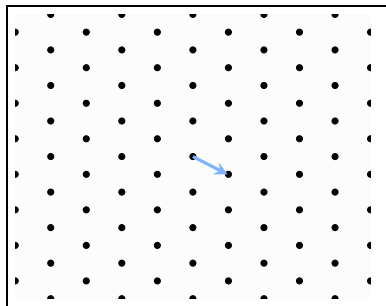
(b) Dual lattice $\mathcal{L}^*$

Lattices

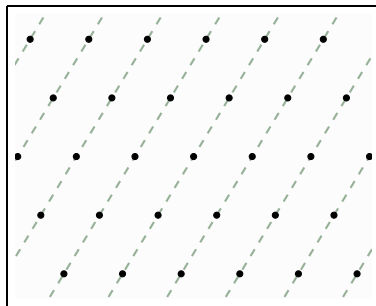
Duality

- Each primal vector $v \in \mathcal{L}^*$ partitions the space of the dual lattice $\mathcal{L}$ in layers ℓ_i orthogonal to v :

$$\ell_i = \{x \in \mathcal{L} : \langle x, v \rangle = i\}$$



(a) Primal lattice $\mathcal{L}$



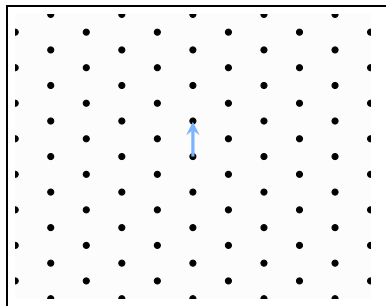
(b) Dual lattice $\mathcal{L}^*$

Lattices

Duality

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(a) Primal lattice $\mathcal{L}$



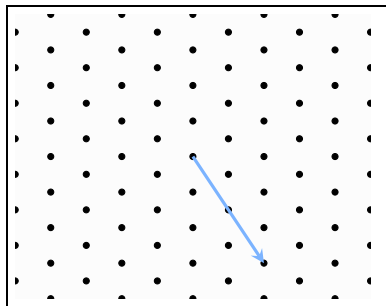
(b) Dual lattice $\mathcal{L}^*$

Lattices

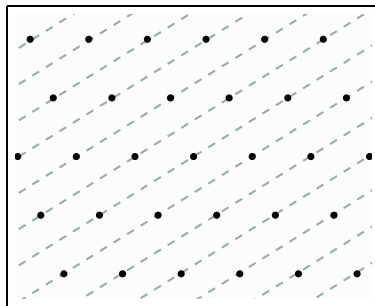
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(a) Primal lattice $\mathcal{L}$

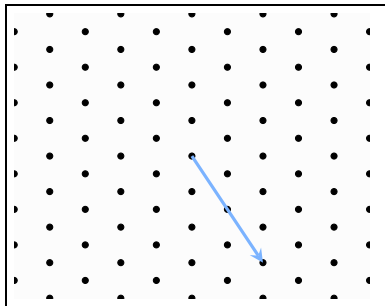


(b) Dual lattice $\mathcal{L}^*$

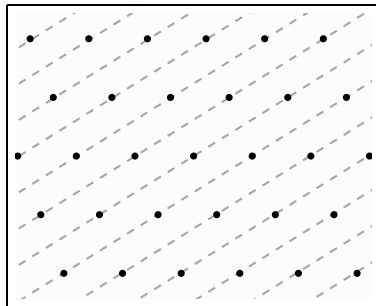
Lattices

Duality

- ▶ Given $\mathbf{v} \in \mathcal{L}^*$, the distance between the partitions is $1/\|\mathbf{v}\|$.
- ▶ The longer the vector in the dual, the “worse” the partitions.



(a) Primal lattice $\mathcal{L}$



(b) Dual lattice $\mathcal{L}^*$

Lattices

Duality

Lemma

$\mathcal{L}_q(A)$ and $\mathcal{L}_q^\perp(A)$ are dual.

Sketch proof

Any point $\mathbf{x} \in \mathcal{L}_q^\perp(A)$ is such that $A^t \mathbf{x} \equiv \mathbf{0} \pmod{q}$. By evaluating an inner product between any point $\mathbf{x}$ of $\mathcal{L}_q^\perp$ and $\mathbf{y} \in \mathcal{L}_q$, we obtain:

$$\langle \mathbf{x}, \mathbf{y} \rangle \equiv \langle \mathbf{x}, A\mathbf{x}' \rangle \equiv \langle A^t \mathbf{x}, \mathbf{x}' \rangle \equiv \langle \mathbf{0}, \mathbf{x}' \rangle \equiv 0 \pmod{q}$$

So they are dual, up to some scaling factor q .

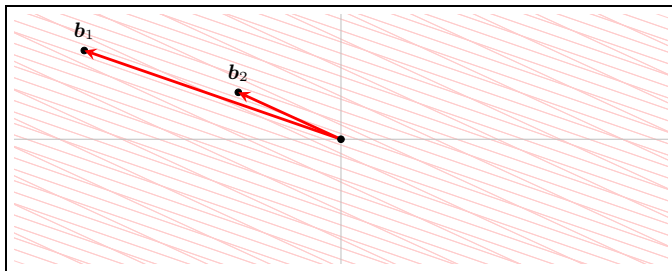


Lattices

Hard problems

Definition (Shortest vector problem (SVP))

Given a lattice $\mathcal{L} \subset \mathbb{R}^n$, find the length λ_1 of the shortest nonzero vector in $\mathcal{L}$.

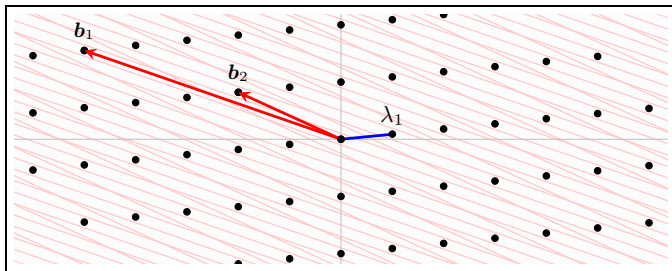


Lattices

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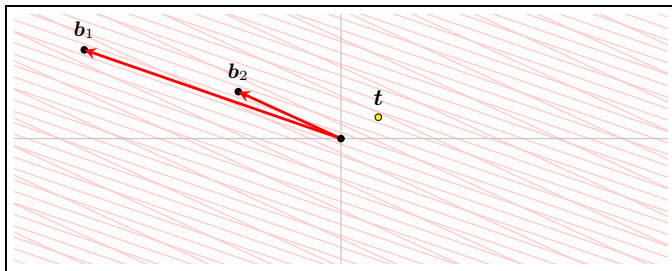


Lattices

Hard problems

Definition (Bounded distance decoding (BDD))

Given $\mathcal{L} \subset \mathbb{R}^n$ and a target $\mathbf{t} \in \mathbb{R}^n$ at distance less than $\frac{1}{2}\lambda_1$ from the lattice, find the closest lattice point to $\mathbf{t}$.

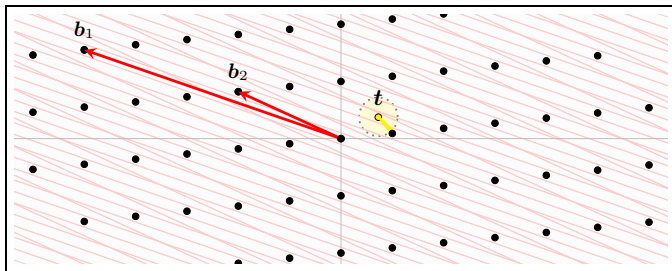


Lattices

Hard problems

Definition (Bounded distance decoding (BDD))

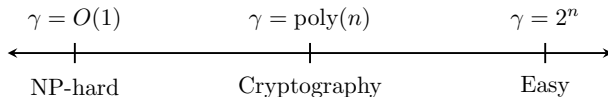
Given $\mathcal{L} \subset \mathbb{R}^n$ and a target $\mathbf{t} \in \mathbb{R}^n$ at distance less than $\frac{1}{2}\lambda_1$ from the lattice, find the closest lattice point to $\mathbf{t}$.



Lattices

Hard problems

- ▶ This problems can also be stated in terms of some approximation factor $\gamma \geq 1$ (e.g., SVP_γ or BDD_γ).
- ▶ For small and constant γ , both are known to be NP-hard [\[?, ?\]](#), for $\gamma = \text{poly}(n)$ we believe them to be very hard (but not NP-hard).



Lattices

Hard problems

- ▶ These are standard computational complexity problems.
- ▶ Usually they are studied in the worst-case.
- ▶ In cryptography, we need average hardness results (e.g. we can not merely use $P \neq NP$).
- ▶ Lattices gained a lot of attention after Ajtai breakthrough [Ajt96], who showed that solving some problems on average is as hard as in the worst case

Lattice problems

Short integer solution

Short integer solution (SIS)

Given a uniformly random $A \in \mathbb{Z}^{m \times n}$, with $m > n \log(q)$ find a nontrivial “short” solution $z \in \mathbb{Z}^m$ such that $A^t z \equiv \mathbf{0} \in \mathbb{Z}^n$ and $z_i \leq \beta$ for all $z_i \in z$.

$$\begin{array}{c} m \\ \hline \boxed{A^t} \\ \hline n \end{array} \cdot \begin{array}{c} \boxed{z} \end{array} = \begin{array}{c} \boxed{0} \end{array} \pmod{q}$$

Lattice problems

Short integer solution

- ▶ Finding a solution is easy (Gaussian elimination) the problem is finding a short one!
- ▶ This allows to define a surjective OWF as

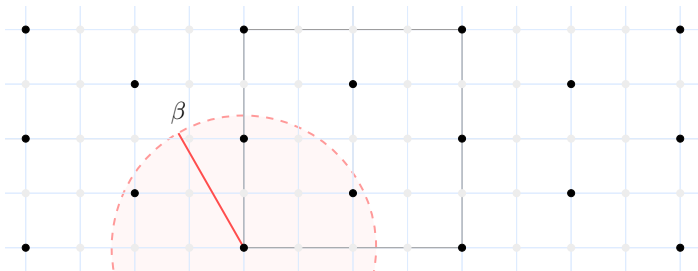
$$f_A(\mathbf{x}) = A^t \mathbf{x} \bmod q : \mathbb{Z}_q^m \rightarrow \mathbb{Z}_q^n$$

- ▶ Useful to build signatures, collision resistant hash functions, etc.

Lattice problems

Short integer solution

- This is “average case” of approximate SVP over the q -ary parity check lattice $\mathcal{L}_q^\perp(A)$!



Lattice problems

Learning with errors

Learning with errors (LWE)

The Learning with errors problem asks to recover a secret vector $\mathbf{s}$ given m noisy linear equations of the form

$$(\mathbf{a}_i, b_i = \langle \mathbf{a}_i, \mathbf{x} \rangle + s_i \pmod{q}),$$

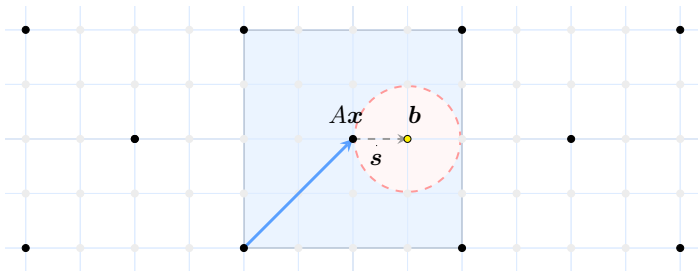
where $\mathbf{a} \in \mathbb{Z}_q^n$ is random, s is small noise, and q is a modulus.

$$\begin{array}{c} n \\ \hline \boxed{A} \\ \hline m \end{array} \cdot \begin{array}{c} \boxed{x} \end{array} + \begin{array}{c} \boxed{s} \end{array} = \begin{array}{c} \boxed{b} \end{array} \pmod{q}$$

Lattice problems

Short integer solution

- This is “average case” of approximate BDD over the row-generated q -ary lattice $\mathcal{L}_q(A)$!



Lattice problems

Short integer solution

- LWE and SIS are very related, and (quantumly [Reg05]) equivalent!

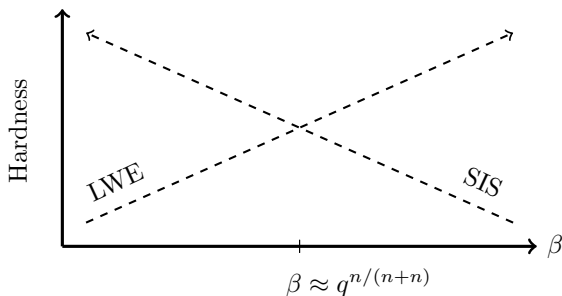


Figure: Figure taken from [Lyu24]

Lattice problems

Learning with errors

Decision-LWE $\leq$ SIS (sketch)

Given an LWE instance $A \in \mathbb{Z}_q^{m \times n}$ and $\mathbf{b} = A\mathbf{x} + \mathbf{s} \in \mathbb{Z}_q^m$, we can build an SIS instance as $A^t \mathbf{z} \equiv \mathbf{0} \in \mathbb{Z}_q^m$. By solving the problem, we find a short $\mathbf{z}$, we can compute

$$\begin{aligned} \mathbf{z}^t \mathbf{b} &\equiv \mathbf{z}^t (A\mathbf{x} + \mathbf{s}) \\ &\equiv \mathbf{z}^t A\mathbf{x} + \mathbf{z}^t \mathbf{s} \\ &\equiv (A^t \mathbf{z})^t \mathbf{x} + \mathbf{z}^t \mathbf{s} \equiv \mathbf{z}^t \mathbf{s} \pmod{q} \end{aligned} \tag{1}$$

If $\mathbf{z}^t \mathbf{s}$ is small (since we know that both $\mathbf{z} \in \mathbb{Z}_q^m$ and $\mathbf{s} \in \mathbb{Z}_q^m$ are small), then $(A, \mathbf{b})$ was almost surely an LWE instance, otherwise it was random noise.

Lattice problems

Learning with errors

Decision-LWE $\leq$ SIS

Given an LWE instance $(A \in \mathbb{Z}_q^{m \times n}, b \in \mathbb{Z}_q^m)$, if we can find $s \in \mathbb{Z}_q^m$ such that $As = b$, we can solve the problem, (2)

If $z^t s$ is small), then $As = b + z^t s$ is close to b (otherwise, it was random noise).



LWE in homomorphic encryption

In order to build a HE scheme, we follow Regev's [Reg05] encryption which uses LWE twice:

- ▶ Key generation: generate a random q -ary basis for $\mathcal{L}_q(A)$. Sample an LWE instance and “plant” it to the basis to obtain a short vector
- ▶ Encryption: safely hide a BDD target in the partitions of dual space and encode the message.

LWE in homomorphic encryption

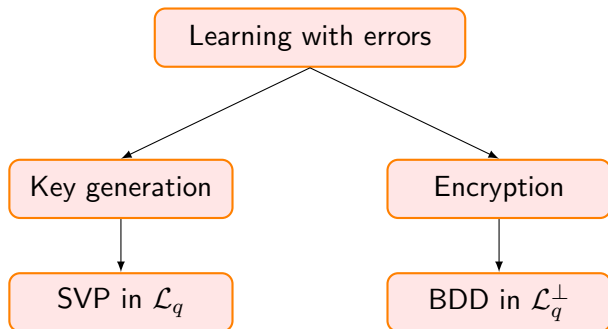


Figure: Usage and lattice interpretation of LWE in an encryption scheme

LWE in homomorphic encryption

Key generation

- ▶ Sample a uniform random lattice basis $B \in \mathbb{Z}_q^{m \times n-1}$.
- ▶ A BDD instance is sampled as:

$$\mathbf{b} \equiv B\mathbf{t} + \mathbf{s} \in \mathbb{Z}_q^m$$

with $\mathbf{t} \in \mathbb{Z}_q^{n-1}$ being a uniform sample and $\mathbf{s} \in \chi^m$ being a “short” error. The secret key is:

$$sk \leftarrow \mathbf{x} \equiv \begin{pmatrix} 1 \\ -\mathbf{t} \end{pmatrix} \in \mathbb{Z}^n$$

- ▶ The vector $\mathbf{b}$ is appended to the basis B to generate the public key pk :

$$pk \leftarrow A \equiv [\mathbf{b} \mid B] \in \mathbb{Z}_q^{m \times n}$$

LWE in homomorphic encryption

Key generation

Let us focus on the public key pk :

$$pk \leftarrow A \equiv (\mathbf{b} \mid B) \in \mathbb{Z}_q^{m \times n}$$

We simply appended $\mathbf{b} \in \mathbb{Z}_q^m$ as a column to the random q -ary basis. As a result, we obtain a (seemingly) random q -ary basis with a planted short vector (i.e., $\mathbf{s}$).

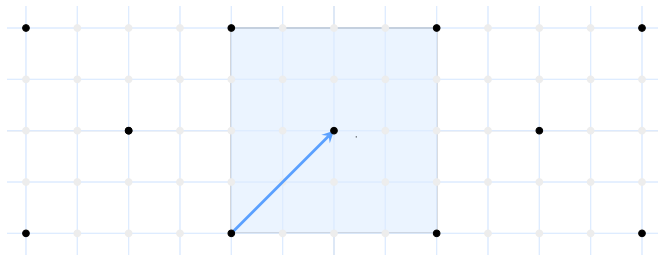
Lattice interpretation

We used LWE to safely inject a short vector in $\mathcal{L}_q(B)$ to obtain $\mathcal{L}_q(A)$.

LWE in homomorphic encryption

Key generation

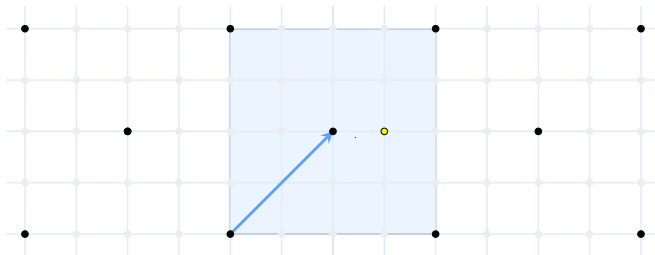
- We start from a random matrix $A \in \mathbb{Z}_q^{m \times n}$. For example fix $m = 2, n = 1, q = 4$.



LWE in homomorphic encryption

Key generation

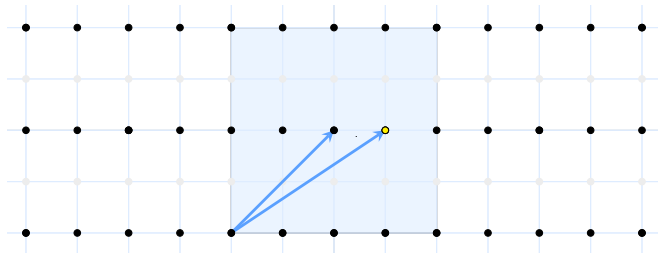
- We sample an LWE instance $\mathbf{b} \in \mathbb{Z}_q^m$



LWE in homomorphic encryption

Key generation

- We append it as part of the basis to create the trapdoor lattice!



LWE in homomorphic encryption

Key generation

How to recover such shortest vector $\mathbf{s} \in \mathbb{Z}_q^m$? Simply use the secret

$$sk \leftarrow \mathbf{x} \equiv \begin{pmatrix} 1 \\ -\mathbf{t} \end{pmatrix} \in \mathbb{Z}_q^n.$$

In particular, the product between pk and sk allows to recover the shortest vector:

$$pk \cdot sk \equiv (\mathbf{b} \mid B) \cdot \begin{pmatrix} 1 \\ -\mathbf{t} \end{pmatrix} \equiv \mathbf{b} - B\mathbf{t} \equiv B\mathbf{t} + \mathbf{s} - B\mathbf{t} \equiv \mathbf{s}$$

LWE in homomorphic encryption

Key generation

- ▶ Remark: short vectors in $\mathcal{L}_q(A)$ can be used to partition the dual lattice
- ▶ Idea: use that “space” to encode the message as a BDD target in the dual space.

LWE in homomorphic encryption

Encryption

How to encrypt a message $\mu \in \mathbb{Z}_t$ ($t \in \mathbb{N}$ is the plaintext space)?

- ▶ Sample a point close to the origin in $\mathcal{L}_q^\perp(A)$ as:

$$A^t \cdot \mathbf{r} \in \mathbb{Z}_q^n \quad \text{with } \mathbf{r} \in \mathbb{Z}_q^m \text{ small}$$

This is a BDD target w.r.t the origin, with error $\mathbf{r}$ – in the dual lattice.

- ▶ Add the message as $A^t \cdot \mathbf{r} + (q/2 \cdot \mu, 0, \dots, 0) \in \mathbb{Z}_q^n$

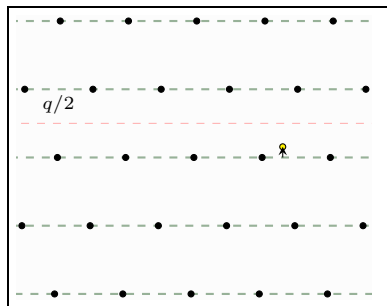
Lattice interpretation

We used LWE to safely hide a BDD target in $\mathcal{L}_q^\perp(A)$.

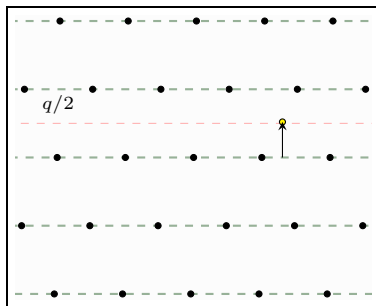
LWE in homomorphic encryption

Encryption

An intuition on how to visualize ciphertexts in $\mathcal{L}_q^\perp(A)$, using the “best” partition given by the shortest $s \in \mathcal{L}_q(A)$:



(a) Encryption of zero



(b) Encryption of one

LWE in homomorphic encryption

Encryption

To recover the message, project it into the “best” partition using the shortest primal vector!

Remark

We can recover the shortest vector $s \in \mathcal{L}_q$ as $pk \cdot sk$.

LWE in homomorphic encryption

Encryption

To recover the message, project it into the “best” partition using the shortest primal vector!

Remark

We can recover the shortest vector $\mathbf{s} \in \mathcal{L}_q$ as $pk \cdot sk$.

Compute the product between the BDD target and the shortest primal vector:

$$\begin{aligned}
 [\langle \mathbf{c}, \mathbf{x} \rangle]_q &\equiv [\langle A^t \cdot \mathbf{r} + (q/2 \cdot \mu, 0, \dots, 0), \mathbf{x} \rangle]_q \\
 &\equiv [\mathbf{r}^t \cdot A \cdot \mathbf{x} + x_1 \cdot q/2 \cdot \mu]_q \\
 &\equiv [\langle \mathbf{r}, \mathbf{s} \rangle + x_1 \cdot q/2 \cdot \mu]_q \\
 &\equiv \varepsilon + q/2 \cdot \mu \text{ (check if small)..}
 \end{aligned}$$

LWE-based encryption

Take home message

Smaller $s \in \mathcal{L}_q$ allow to handle large errors $r \in \mathbb{Z}_q^n$ since:

$$|\langle r, s \rangle| \leq \|r\| \cdot \|s\|$$

LWE-based encryption

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- We also connect to the geometric intuition – remark that the width of the partitions is $1/\|s\|$.

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Take home message

Smaller $s \in \mathcal{L}_q$ allow to handle large errors $r \in \mathbb{Z}_q^n$ since:

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- ▶ We also connect to the geometric intuition – remark that the width of the partitions is $1/\|s\|$.
- ▶ Spoiler: for multiplications we will need to handle **very large** errors...

LWE-based encryption

BGV-like schemes

- This is the high-level idea behind the BGV [BGV12] scheme, and other following works [Bra12,FV12,CKKS17].

Homomorphic Encryption for Arithmetic of Approximate Numbers

Jung Hee Cheon¹, Andrey Kim¹, Miran Kim², and Yongsoo Song¹

Somewhat Practical Fully Homomorphic Encryption *

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Abstract. In this paper we port on the Learning With Errors (LWE) to introduce two optimised versions of the scheme. The first version uses a smaller relinearisation key, but it

Fully Homomorphic Encryption without Bootstrapping

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Weizmann Institute of Science

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IBM T.J. Watson Research Center

Vinod Vaikuntanathan[†]
University of Toronto

Abstract. We summarize arithmetic. In addition, together with a procedure truncates a result. The main idea is to add noise during approximate result, our decryption precision. We also propose a new algorithm. A polynomial is an element of a vector of complex

LWE-based encryption

BGV-like schemes

► Additions:

$$[[\langle \mathbf{c}_1 + \mathbf{c}_2, \mathbf{x} \rangle]_q]_2 \equiv [[\langle \mathbf{c}_1, \mathbf{x} \rangle + \langle \mathbf{c}_2, \mathbf{x} \rangle]_q]_2$$

► Multiplications:

$$[[\langle \mathbf{c}_1 \otimes \mathbf{c}_2, \mathbf{x} \otimes \mathbf{x} \rangle]_q]_2 \equiv [[\langle \mathbf{c}_1, \mathbf{x} \rangle \cdot \langle \mathbf{c}_2, \mathbf{x} \rangle]_q]_2$$

- For compactness, we need a *key switching* operation (given a ciphertext under some key $\mathbf{x}_1 \in \mathbb{Z}_q^n$, convert it into a ciphertext under another key $\mathbf{x}_2 \in \mathbb{Z}_q^m$ that decrypts to the same message).

LWE-based encryption

BGV-like schemes

Sketch on key switching

Given two keys $s_1 \in \mathbb{Z}_q^{n_1}$, $s_2 \in \mathbb{Z}_q^{n_2}$, define a *key switching key* as

$$A_{ks} \equiv (b_2 \mid A') \in \mathbb{Z}_q^{n_1 \times n_2}$$

Then, add column-wise in the first column the old key s_1 to obtain

$$A_{ks} \equiv (b_2 + s_1 \mid A') \in \mathbb{Z}_q^{n_1 \times n_2}$$

And multiply the old ciphertext $c \in \mathbb{Z}_q^{n_1}$ to obtain a new ciphertext under the new key (and dimension)

$$c^t \cdot A_{ks} \equiv c^t \cdot (b_2 + s_1 \mid A') \equiv (b_2 \mid A') \cdot (r + m) \in \mathbb{Z}_q^{n_2}$$

(many simplifications done, ignoring bit decompositions, but this is the main idea)

LWE-based encryption

Ring setting

- ▶ Until now, we described the LWE setting.
- ▶ Nevertheless, (almost) everything can be also re-stated for Ring-LWE.
- ▶ Remark that in RLWE the public key is a random polynomial $a(X) \in \mathbb{Z}_q[X]/(X^N + 1)$.
- ▶ This family of polynomial refers to the special case of anti-circulant matrices, which generate a specific type of lattice, called *ideal lattices*.

LWE-based encryption

Ring setting

Toy example (part 1)

Given the polynomial ring $\mathbb{Z}[X]$, define its quotient ring $\mathbb{Z}[X]/p(X)$. We pick a very specific polynomial $p(X)$, called cyclotomic polynomial. So we fix

$$\mathbb{Z}[X]/(X^n + 1)$$

Under this setup, $X^n = -1$. If we fix, e.g. $n = 4$, and a (random) polynomial

$$a(X) = 3 + 1X - 4X^2 + 5X^3 \in \mathbb{Z}[X]/(X^4 + 1)$$

and consider its coefficients $(3, 1, -4, 5)$ as a basis column vector, we can obtain all the others by multiplying by all the powers of X .

LWE-based encryption

Ring setting

Toy example (part 2)

So we obtain

$$3 + 1x - 4X^2 + 5X^3 \cdot X = -5 + 3X + 1X^2 - 4X^3$$

so another vector is $(-5, 3, 1, -4)$, and so on. The final basis obtained by $a(X)$ is

$$\begin{pmatrix} 3 & 1 & -4 & 5 \\ -5 & 3 & 1 & -4 \\ 4 & -5 & 3 & 1 \\ -1 & 4 & -5 & 3 \end{pmatrix}$$

considering rows as basis vectors.

Requirements for HE

Abstracting away from LWE

- ▶ In [LRvW25], we tried to answer the question:
Which are the minimum lattice requirements to build a HE scheme?
- ▶ Nowadays, (R)LWE is almost used as a black box for most constructions.
- ▶ We lose the lattice perspective! Is LWE the only assumption that can be used to build such a scheme?

Requirements for HE

Abstracting away from LWE

It turns out that the minimum requirements are:

- ▶ A lattice $\mathcal{L}$ with an exceptionally short vector s (the shorter, the larger the partitions, the more the operations that we can evaluate).

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It turns out that the minimum requirements are:

- ▶ A lattice $\mathcal{L}$ with an exceptionally short vector s (the shorter, the larger the partitions, the more the operations that we can evaluate).
- ▶ An ambient space where decoding BDD targets is hard (in LWE, decoding BDD targets it is hard by LWE assumption).
- ▶ An ambient space where we can sample small BDD targets (in LWE we have $\mathbb{Z}^m$, by definition of q -ary lattices).

Requirements for HE

Abstracting away from LWE

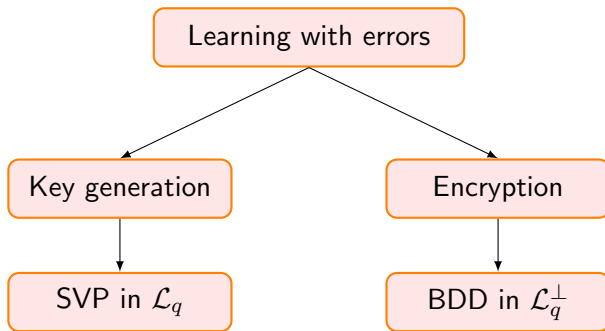


Figure: Usage and lattice interpretation of LWE in an encryption scheme

Requirements for HE

Abstracting away from LWE

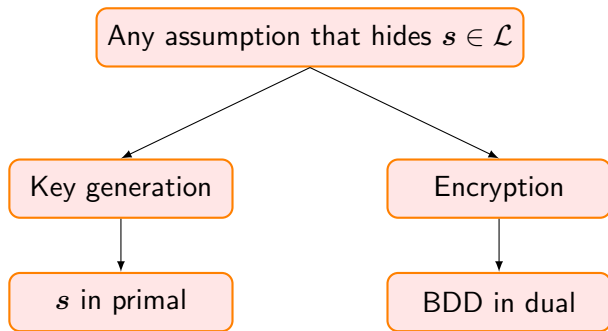


Figure: Generalizing the role of lattices in a FHE scheme

Requirements for HE

Lattice isomorphism problem

- Idea: can we use other assumptions other than LWE?

Requirements for HE

Lattice isomorphism problem

- Idea: can we use other assumptions other than LWE?

Definition (Lattice isomorphism problem (LIP))

Given a pair of isomorphic lattices $\mathcal{L}(B)$ and $\mathcal{L}(B')$, where $B' = O \cdot B \cdot U \in \text{GL}_n(\mathbb{R})$, recover either $O \in \mathcal{O}_n(\mathbb{R})$ or $U \in \text{GL}_n(\mathbb{Z})$

- Remark that two lattices are equivalent if their bases differ by a unimodular matrix.

Requirements for HE

Lattice isomorphism problem

- ▶ LIP gives us a nice framework that allows to work with any lattice, not only q -ary lattices.
- ▶ For instance, we will work in (a rotation of) the trivial $\mathbb{Z}^n$ lattice
- ▶ The LIP instance will simply be $B' = O \cdot U$, i.e., a rotation of $\mathbb{Z}^n$.

Sketch of LIP-BFV

Key generation

We give a sketch of a possible LIP-based instantiation of the BGV scheme.

- Keygen: generate a LIP instance, and set

$$(pk, sk) = (O \cdot U \in \text{GL}_n(\mathbb{R}), O \in \text{GL}_n(\mathbb{R}))$$

Lattice interpretation

The public key is a bad lattice basis. The secret key is the set of (rotated) unit vectors of $\mathbb{Z}^n$. Each $\mathbf{o}_i$ is a short vector.

Sketch of LIP-BFV

Encryption

We give a sketch of a possible LIP-based instantiation of the BFV scheme.

- Encryption: generate a BDD target in the dual space, and add the message

$$\mathbf{c} \equiv \mathcal{D}_{\mathcal{L}(O.U)^{-1}, \sigma} + (q/2 \cdot \mu, 0, \dots, 0) \in \mathbb{R}^n$$

Lattice interpretation

An encryption is a BDD target in $\mathcal{L}^*$

Sketch of LIP-BFV

Encryption

Warning

In an actual instantiation, we can not sample easily as $\mathcal{D}_{\mathcal{L}(O.U)^{-1}, \sigma}$, but we will use a trick presented in Section 3.3 of [LRvW25]

Extra: bootstrapping

- ▶ Until now, we talked about HE, how to obtain **fully** homomorphic encryption?
- ▶ Theoretically, we could set parameters to allow the computation of any circuit.
- ▶ Unpractical for deep circuits.

Extra: bootstrapping

- ▶ Until now, we talked about HE, how to obtain **fully** homomorphic encryption?
- ▶ Theoretically, we could set parameters to allow the computation of any circuit.
- ▶ Unpractical for deep circuits.
- ▶ Solution: **bootstrapping**.

Bootstrapping [Gen09]

Given a ciphertext c with some amount of noise n , return c' with noise $n' < n$ by homomorphically evaluating the decryption circuit using $\text{enc}(sk)$.